

GRIVITA, Romania

WMO: 154050

Lat: 44.7408N

Lon: 27.2947E

Elev: 52

StdP: 100.71

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.4	(c) -10.0	(d) -14.4	(e) 1.1	(f) -11.6	(g) -12.0	(h) 1.3	(i) -8.8	(j) 17.4	(k) -5.3	(l) 14.4	(m) -4.3	(n) 3.3	(o) 230	(p) 0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.0	(c) 33.5	(d) 22.1	(e) 32.1	(f) 21.3	(g) 30.8	(h) 20.9	(i) 23.5	(j) 30.8	(k) 22.7	(l) 29.7	(m) 21.8	(n) 28.5	(o) 2.8	(p) 50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.2	26.2	20.5	15.3	25.2	19.7	14.5	24.4	70.4	30.6	66.9	30.0	64.0	28.4	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.1	(b) 10.0	(c) 8.3	(e) -16.3	(f) 35.9	(g) 2.4	(h) 1.8	(i) -18.0	(j) 37.2	(k) -19.4	(l) 38.2	(m) -20.7	(n) 39.3	(o) -22.4	(p) 40.6		
(a) 12.1	(b) 10.0	(c) 8.3	(e) -16.6	(f) 25.5	(g) 2.3	(h) 1.7	(i) -18.2	(j) 26.8	(k) -19.6	(l) 27.8	(m) -20.9	(n) 28.7	(o) -22.6	(p) 30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	-1.5	2.0	6.0	11.6	17.2	21.7	23.1	18.5	11.9	5.9	0.4	(6)	
(7)		DBStd	9.56	4.51	4.97	4.63	4.11	3.37	2.78	2.47	3.00	4.20	4.14	4.71	4.63	(7)
(8)		HDD10.0	1218	358	226	137	31	1	0	0	0	28	138	299		(8)
(9)		HDD18.3	2913	616	457	382	204	61	6	1	3	52	204	372	557	(9)
(10)		CDD10.0	1838	0	2	13	78	224	351	407	404	256	86	17	0	(10)
(11)		CDD18.3	490	0	0	0	1	25	106	149	149	57	3	0	0	(11)
(12)		CDH23.3	5038	0	0	2	43	321	1041	1493	1548	551	39	1	0	(12)
(13)		CDH26.7	1912	0	0	0	7	77	363	599	678	181	7	0	0	(13)
(14)	Wind	WSAvg	3.2	3.6	3.6	4.0	3.6	3.1	2.7	2.6	2.9	2.9	3.0	3.6	(14)	
(15)	Precipitation	PrecAvg	496	31	20	31	33	58	68	46	50	40	35	40	(15)	
(16)		PrecMax	710	84	56	72	91	156	115	110	97	147	79	104	112	(16)
(17)		PrecMin	320	3	1	0	4	11	13	7	9	1	3	0	3	(17)
(18)		PrecStd	101	20	14	19	17	36	25	26	25	35	22	24	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	18.4	22.4	27.5	31.7	33.6	35.1	35.2	32.7	27.6	21.5	12.6	(19)
(20)			MCWB	7.9	12.6	13.4	17.7	20.5	22.8	23.5	23.3	19.8	19.2	14.8	10.2	(20)
(21)		2%	DB	7.6	14.2	19.0	24.1	28.6	31.7	33.3	33.6	30.5	23.8	18.0	10.1	(21)
(22)			MCWB	6.1	10.2	11.6	15.7	19.2	21.6	22.3	22.0	19.3	16.8	13.7	8.3	(22)
(23)		5%	DB	5.7	11.4	16.3	21.8	26.7	30.4	31.8	32.2	28.7	21.3	14.8	7.9	(23)
(24)			MCWB	4.8	8.2	10.2	14.4	18.1	21.2	21.6	21.3	18.6	15.3	11.7	6.6	(24)
(25)		10%	DB	3.8	8.6	13.6	19.8	24.8	28.9	30.1	30.7	26.7	18.9	12.7	5.9	(25)
(26)			MCWB	3.1	6.7	8.7	13.3	16.8	20.6	21.0	20.8	18.0	14.3	10.8	4.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	12.9	14.2	18.5	21.3	24.1	24.5	25.0	21.7	20.0	15.4	11.2	(27)
(28)			MCD B	9.5	17.7	21.3	26.1	29.1	31.4	32.1	32.5	28.5	26.0	19.9	12.7	(28)
(29)		2%	WB	6.4	10.4	12.1	16.5	20.0	22.9	23.5	23.5	20.7	17.5	13.7	8.4	(29)
(30)			MCD B	7.4	14.0	17.8	22.6	27.4	29.6	30.6	31.2	26.9	22.1	17.1	9.9	(30)
(31)		5%	WB	4.9	8.5	10.7	15.1	18.9	22.0	22.6	22.5	19.8	16.1	12.3	6.7	(31)
(32)			MCD B	5.7	10.8	15.6	20.8	25.0	28.9	29.4	29.8	25.7	20.3	14.4	8.1	(32)
(33)		10%	WB	3.2	7.0	9.2	13.9	17.7	21.2	21.8	21.5	18.8	14.8	10.9	4.9	(33)
(34)			MCD B	3.8	8.6	13.0	18.7	23.3	27.5	28.0	28.3	25.0	18.2	12.4	5.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	7.6	9.9	11.8	12.3	11.9	12.0	12.3	12.1	10.3	7.3	6.1	(35)
(36)			MCDBR	7.8	13.0	15.4	15.9	15.0	14.3	14.9	14.6	15.0	14.2	11.8	8.9	(36)
(37)			MCWBR	6.1	8.7	9.0	8.3	6.8	5.7	5.7	5.9	6.0	7.4	7.2	6.8	(37)
(38)		5% WB	MCD BR	7.5	12.0	14.6	14.7	14.2	13.4	12.8	12.9	13.4	12.8	10.4	8.8	(38)
(39)			MCWBR	6.0	8.4	8.8	8.3	7.0	5.7	5.5	6.1	5.9	7.2	6.7	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.344	0.373	0.406	0.461	0.434	0.437	0.451	0.465	0.430	0.392	0.366	0.346	(40)	
(41)		taud	2.296	2.221	2.182	2.042	2.203	2.226	2.195	2.145	2.206	2.298	2.320	2.312	(41)	
(42)		Ebn at Noon	749	792	818	801	839	836	818	789	784	763	715	704	(42)	
(43)		Edh at Noon	84	108	128	160	140	138	141	143	124	99	82	75	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.33	2.15	3.34	4.67	5.89	6.33	6.42	5.75	4.18	2.67	1.47	1.12	(44)	
(45)		RadStd	0.19	0.31	0.35	0.48	0.42	0.43	0.40	0.33	0.31	0.30	0.17	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	-123	+70	+44

Nomenclature: See separate page